3 A bungee jumper of mass 64 kg secures one end of an elastic rope to a bridge. The other end is attached to the bungee jumper. The jumper falls from rest from the bridge and descends into the valley below, as shown in Figure 3.1.

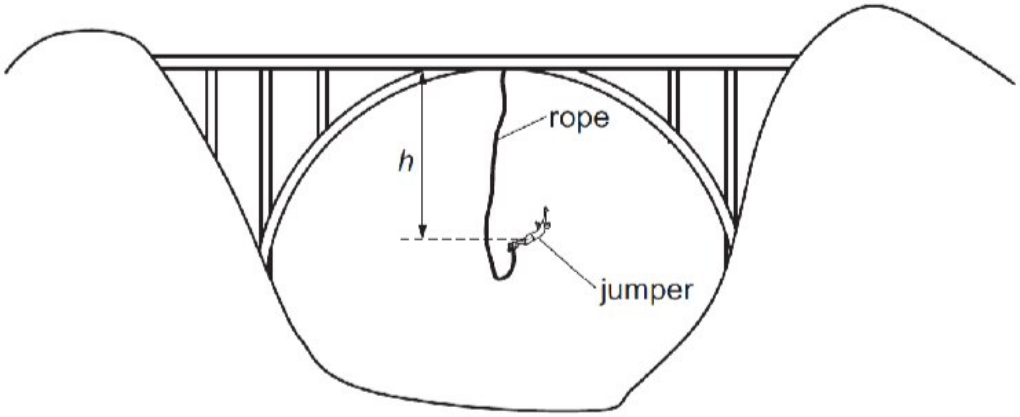



**Figure 3.1** (not to scale)

Figure 3.2 shows the variation of the tension $T$ in the rope with the vertical distance $h$ of the jumper below the level of the bridge.

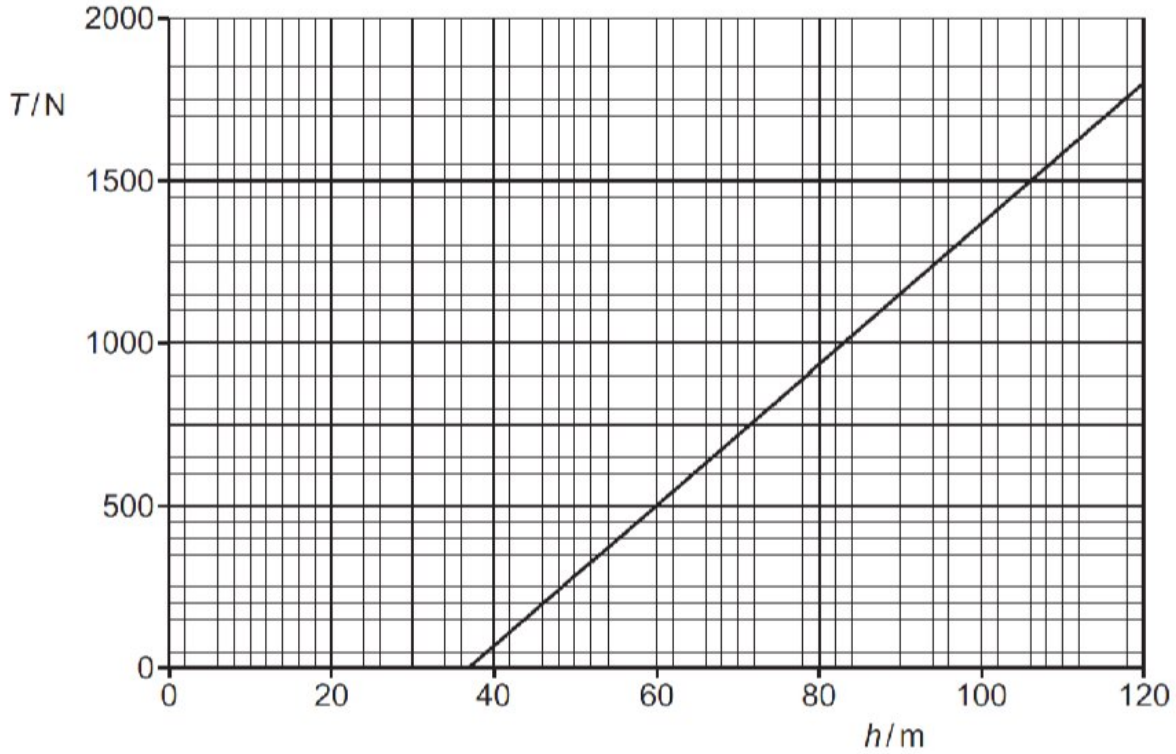



**Figure 3.2**

**(a)** The position of the bungee jumper is at a distance of 120 m below the bridge.

**(i)** Show that the loss of gravitational potential energy since leaving the bridge is 75 kJ.

[1]

**(ii)** Calculate the elastic potential energy in the rope.

elastic potential energy = .....J [2]

**(b)** Explain what can be deduced from the information in **3(a)** about the speed of the bungee jumper when at a distance of 120 m below the bridge.

.....

.....

..... [2]

[Total: 5]